Physics of the Earth and Planetary Interiors, 64 (1990) 303–313
Elsevier Science Publishers B.V., Amsterdam

The palaeomagnetism of the Tumblagooda Sandstone, Western Australia: Gondwana palaeozoic apparent polar wandering

P.W. Schmidt and B.J.J. Embleton

CSIRO Division of Exploration Geoscience, PO Box 136, North Ryde, N.S.W. 2113 (Australia)

(Accepted for publication February 20, 1989)

ABSTRACT

Schmidt, P.W. and Embleton, B.J.J. The palaeomagnetism of the Tumblagooda Sandstone, Western Australia: Gondwana palaeozoic apparent polar wandering. Phys. Earth Planet. Inter., 64: 303–313.

A magnetic polarity reversal near the top of the Tumblagooda Sandstone, in the Carnarvon Basin, Western Australia, has been intensively investigated in five coastal sections and four inland sections, all located near the top of the formation. A further four inland sections towards the base of the formation were also sampled. Of the inland sections, all but two are suspected to have been largely remagnetized in similar fashion to Late Palaeozoic and Mesozoic sediments to the south.

At the remaining seven sections a zone of normal magnetization overlies a zone of mixed (coexisting normal and reversed) magnetization, and the boundary between them coincides with a unique marker horizon of pebbly sandstone. The intimate relationship of the (virtual) reversal and the pebble marker horizon lends strong support to the suggestion that erosional stripping accompanied the formation of the pebble marker. The pebble marker is best interpreted as a lag deposit. The stratigraphic coincidence does not necessarily signify the concurrence of a reversal with the formation of the marker horizon, but rather that the relationship represents a variation on the unconformity test. The record of the actual geomagnetic reversal has been physically removed from most sections.

These results show conclusively that the age of the magnetization of the Tumblagooda Sandstone is very close to the age of deposition, and can therefore provide an important pole for the definition of the Gondwana apparent polar wander path. However, the suggested Silurian age must be questioned on the basis of the palaeomagnetism which best accords with an Ordovician age. The lower parts of the sequence may be as old as the Cambrian. While the age of the palaeomagnetic pole from the Tumblagooda Sandstone is therefore not well constrained, its position is well defined and in combination with other Gondwana poles indicates a period of rapid polar motion during the Ordovician/Silurian. A collision with Laurasia at the end of the Silurian and beginning of the Devonian may explain the timing of the Caledonian/Acadian Orogenies.

1. Introduction

An earlier palaeomagnetic investigation (Embleton and Giddings, 1974) of the Tumblagooda Sandstone of Western Australia detected the presence of a reversal in polarity of the remanent magnetization. This formation is a prominent red bed unit outcropping in gorges along the coast and banks of the Murchison River. Of seventeen samples, eleven from the upper two thirds of the 20-m section displayed normal polarity, while six samples from the lower third of the section displayed reversed polarity. The mean direction, after taking the anti-directions of the reversed directions, was found to be $D = 65°$, $I = -33°$ and $\alpha_{95} = 9°$. The reversal apparently occurred over a thickness of < 2 m. We perceived this to be an ideal opportunity to document a polarity reversal in a red bed sequence.

Many recent studies of Palaeozoic continental reconstructions, based on palaeomagnetic results, have served to emphasise the ambiguities present in the earlier data set (Kent et al., 1984; Livermore et al., 1985; Schmidt et al., 1986; Hurley and

0031-9201/90/$03.50 © 1990 – Elsevier Science Publishers B.V.

Van der Voo, 1987; Schmidt et al., 1987; Hargraves et al., 1987; Van der Voo, 1988; Miller and Kent, 1988). Many rocks have failed to yield the quality of data necessary to constrain and differentiate the different models that have been proposed. Although much of the Gondwana data have been derived from Australian rocks, the autochthonous (in situ) versus the allochthonous (exotic/displaced) models for the Lachlan Fold Belt of eastern Australia remains a source of uncertainty. Data from Western Australia are therefore paramount to Palaeozoic continental distributions. However, the trade-off between eastern and western Australia as study areas of Palaeozoic rocks, is the minor folding of strata, and dearth of igneous rocks in the west. The lack of folding makes validation of the age of magnetization difficult while the lack of igneous rocks confines work to sedimentary rocks. The irony is that, to be removed from the suspect terrain syndrome, a new set of problems have to be addressed.

We have resampled the Tumblagooda Sandstone more extensively to investigate the exposed sequence more thoroughly and to document the palaeomagnetic result using the more refined analytical techniques now available. Since the work of Embleton and Giddings (1974) showed that there was little directional change on thermal cleaning, the polarity of the characteristic magnetization of these red beds is reflected by the uncleaned natural remanent magnetization (NRM) measurements. A portable fluxgate magnetometer could therefore be used in the field to give polarity information and greatly expedite sampling.

2. Geology and sampling

Several workers have examined the Tumblagooda Sandstone (Konecki et al., 1958; Condon, 1965; Hocking, 1978; Hocking, 1981; Hocking et al., 1987) which crops out over an area of about 7000 km^2 along the central coast of Western Australia. It is a thick (> 1800 m) sequence of red to buff fluviatile, tidal and coastal sediments. Dating continental redbeds is notoriously difficult but the age of the Tumblagooda Sandstone is thought to be Silurian (R. Hocking, personal communication, 1987; Hocking et al., 1987), at least in part, since it is conformably overlain by the Upper Silurian Dirk Hartog Formation. Hocking also states that although the Tumblagooda Sandstone is very thick, its age is unlikely to extend into the Ordovician because mineralization in adjoining dolerite dykes is dated at 435 ± 15 Ma and there is no indication of dolerite intrusion, mineralization, or the prior passage of mineralizing fluids into the sandstone.

The type section dips 5° or less to the west and is continuously exposed, extending about 70 km along the lower Murchison River gorge. Outcrops in the coastal cliffs adjoin the type section. Marker horizons have been recognized (Hocking, 1981) and one in particular appeared to be closely associated with the occurrence of the magnetic polarity reversal as originally discovered by Embleton and Giddings (1974). This horizon comprises a distinctive pebbly, sandstone bed about 1 m thick, and is well exposed in the coastal cliffs. The bed is composed of ill-sorted rounded quartz and feldspar pebbles set in a coarse sandstone matrix. Five coastal sections spread over 5 km to the south of Kalbarri have been sampled, each spanning both the pebble marker and the polarity reversal. These sections are at Red Bluff, Rainbow Valley, two at Eagle Gorge and another at Goat Gulch (Fig. 1). At each section the position of the polarity reversal was found using the portable fluxgate magnetometer.

A further four sections were sampled several kilometres (5–8 km) to the north-northeast of Kalbarri, near the top of the type section, where a distinctive pebble marker has been correlated with the marker in the coastal sections (Hocking, 1981). These sections are at Thiridine Point, Stonewall, Stonewall North (Fig. 1) and Larrys Soak. Four more sections were collected from along the gorge of the Murchison River, where lower units of the Tumblagooda Sandstone are exposed. In descending stratigraphic position (to the east) these sections outcrop at The Loop, Hawks Head, the Stream Gauging Station and at the Hardabut Fault. The last section was sampled not only because it comprises the lowest units available in the outcrop, but also because the beds here dip 20–25° to the southeast (N125° E) having been disturbed by movement along the fault. This was to enable a

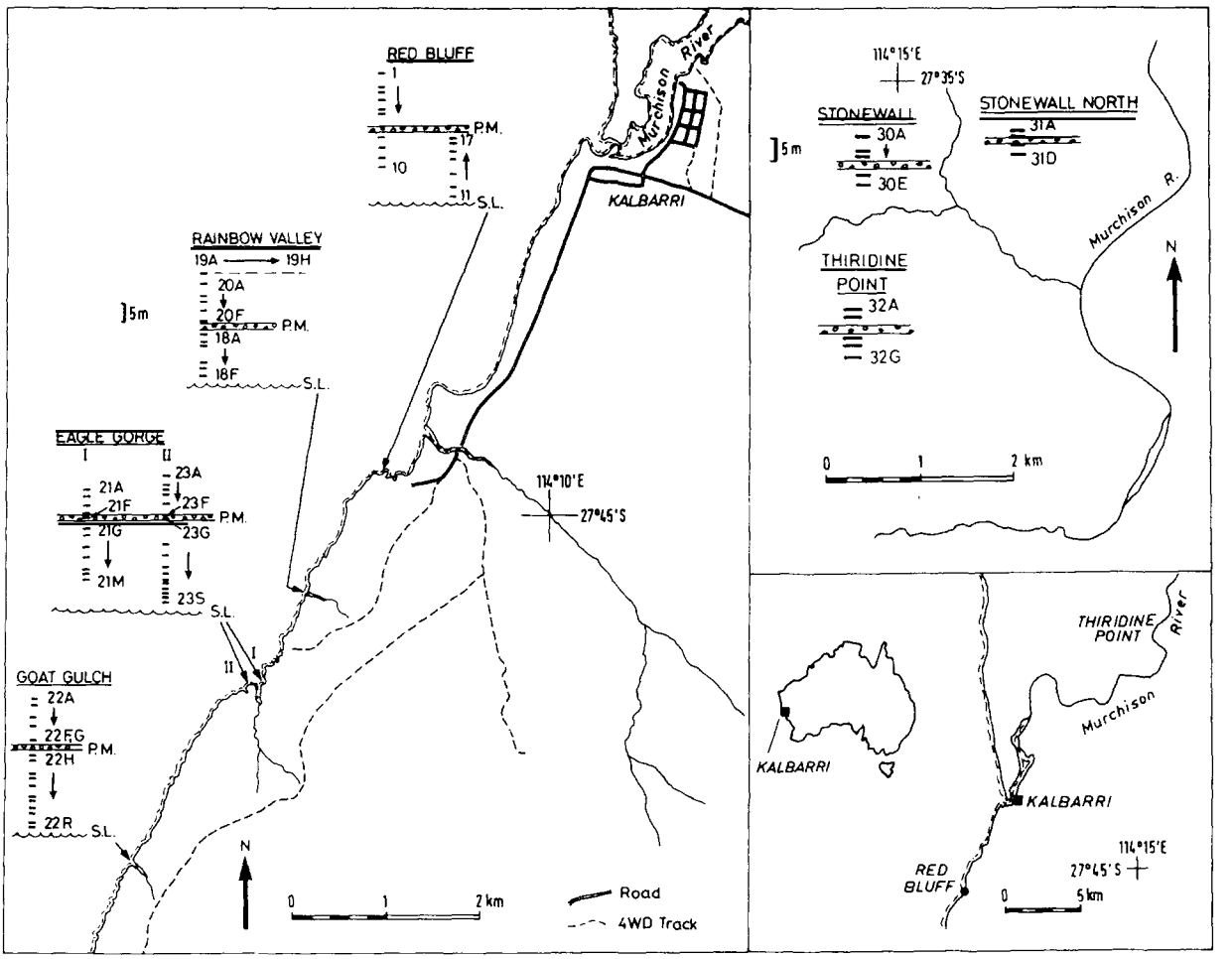


Fig. 1. Location map of sections sampled. P.M. indicates the stratigraphic position of a well-defined pebble marker horizon, while S.L. indicates approximate sea level.

fold test (Graham, 1949; McFadden and Jones, 1981) to be carried out. Overall 220 oriented samples were collected using a magnetic compass, in conjunction with a sun compass when weather permitted.

3. Techniques and results

In the laboratory, magnetizations were measured using a 3-axis CTF cryogenic magnetometer housed inside three orthogonal 4-m^2 Helmholtz coil sets. The coils afford a high degree of control over the measurement environment and prevent specimens from acquiring spurious magnetic components after cleaning. A three stage automated non-magnetic furnace was used for step-wise thermal demagnetization. Orthogonal projections (Zijderveld, 1967) and Linearity Spectrum Analysis (Schmidt, 1982) were used to analyse the magnetic components. The NRM measurements performed in the field were confirmed in the laboratory.

Selected specimens were thermally demagnetized in steps of 200, 300, 400, 500, 550, 590, 630, 670 and 690 ° C. Representative demagnetization trends are displayed in Fig. 2 as orthogonal projections. There is a clear definition between

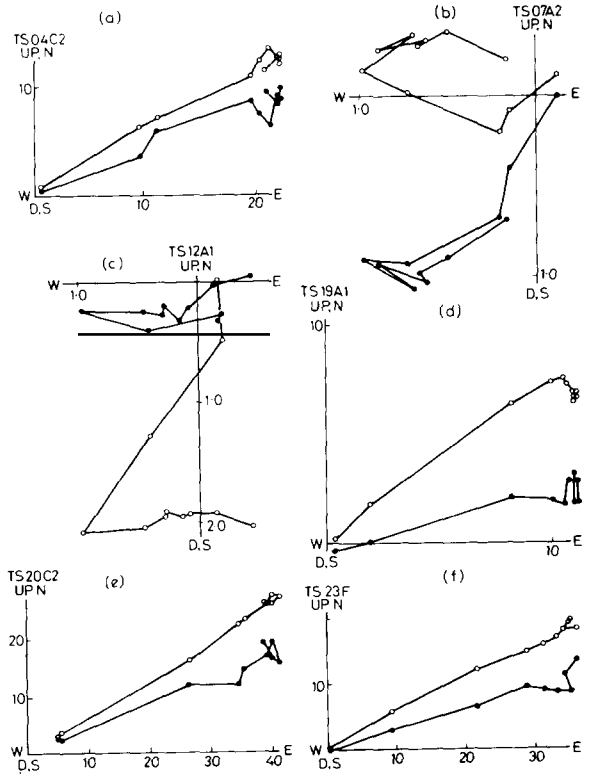


Fig. 2. Orthogonal projections of thermal demagnetization results. Steps following the NRM are 200, 300, 400, 500, 550, 590, 630, 670 and 690°C. Closed (open) symbols refer to the horizontal (vertical) plane. Units are in mA m^{-1}. Samples above (TS04C2, TS19A1 and TS20C2) and within (TS23F) the pebble marker carry a normal component of magnetization, while samples below the marker (TS07A2 and TS12A1) carry a complex component that is not well resolved on thermal or chemical demagnetization (see text).

results from samples below the marker horizon on the one hand, and from samples above and within the marker horizon on the other. Figure 2a (specimen TS04C2 from above the marker horizon at Red Bluff) changes direction very little and is considered to reflect a single, magnetic component directed to the east-northeast shallow up. Indeed most samples from the coastal cliff sections are well behaved on cleaning, their NRM directions being largely free of viscous and other young magnetizations. Other samples e.g. Fig. 2b (specimen TS07A2 from immediately below the marker horizon at Red Bluff), reveal a complex structure that is not well resolved by thermal demagnetization, but appears to be a northeasterly directed component superimposed on a southwesterly directed component. A similar structure is again seen in specimen TS12A1, from well below the pebble marker at Red Bluff (Fig. 2c). The unblocking temperature spectra of these complex magnetic components are, in general, overlapped giving rise to very erratic demagnetization behaviour.

From another locality (Rainbow Valley) 2 km south of Red Bluff (Fig. 1), specimens from above the pebble horizon are seen to be simply magnetized in a WSW–ENE direction (specimens TS19A1 and TS20C2, Fig. 2d and e). Further south at Eagle Gorge II, specimen TS23F from within the pebble marker is also seen to be simply magnetized. Samples from within the pebble marker were confined to the least coarse bands of the matrix. Generally, there is a dichotomy between simple and normal magnetizations within and above the pebble marker, and complex (mostly reversed but with some normal overprinting) magnetizations from below the marker.

Since the magnetizations of samples from below the pebble marker could not be resolved with sufficient accuracy using thermal demagnetization, detailed chemical demagnetization experiments (Collinson, 1974; Roy and Park, 1974) were performed using various strength solutions of hydrochloric acid, but proved to be no more successful. For these experiments standard samples were sliced into thin discs to allow for more even acid leaching and enhanced resolution. Nevertheless, directions of magnetization did not change significantly and samples that displayed intermediate NRM directions typically remained steadfast during the whole leaching process.

To compare these results with those obtained earlier (Embleton and Giddings, 1974) we have taken all thermal demagnetization results from within and above the marker horizon, that is from magnetizations composed entirely of one component, and applied Linearity Spectrum Analysis (Schmidt, 1982) to each section. Magnetizations from below the marker horizon, while largely of reversed polarity, are contaminated by small to moderate components of normal magnetization which could not be preferentially demagnetized

TABLE 1

Summary of palaeomagnetic results

Locality	N [a]	R [a]	Dec (°)	Inc (°)	α_{95}°
Red Bluff	8	7.683	69.7	−34.6	12
Rainbow Valley	6	5.800	72.5	−37.0	14
Eagle Gorge I	5	4.874	71.4	−34.6	14
Eagle Gorge II	5	4.801	68.8	−37.3	17
Goat Gulch	6	5.863	68.5	−40.8	11
Thiridine Pt./ Stonewall	4	3.908	69.6	−36.1	16
Mean	6	5.991	69.6	−36.1	3

[a] N is the number of unit directions and R their resultant, from thermally cleaned specimens. Dec is the declination and Inc the inclination and α_{95} is the half-angle of the cone of 95 percent confidence (Fisher, 1953).

Pole position. Lat = 26.7°S, Long = 33.7°E ($d_p = 2°$, $d_m = 3°$).

and they cannot therefore be used in the calculation of the mean direction. The results are given in Table 1 which shows a mean direction of $D = 69.6°$, $I = -36.1°$, $\alpha_{95} = 3°$, and a pole position of lat = 26.7°S, long = 33.7°E ($d_p = 2°$, $d_m = 3°$) which is not significantly different from, but much more precise than, the earlier results of $D = 65°$, $I = -33°$, $\alpha_{95} = 9°$, and lat = 30°S, long = 31°E, $\alpha_{95} = 9°$.

Results from the other sections sampled are affected variously by Tertiary weathering, particularly the lower sections well inland up the Murchison Gorge (at The Loop, Hawks Head, the Stream Station and at the Hardabut Fault). This weathering is likely to be of a similar age to that found in the north Perth Basin where sediments ranging from Permian to Cretaceous in age have been completely remagnetized by lateritization in the Miocene (Schmidt and Embleton, 1976) The fold-test proposed during sampling could not be performed. Of the remaining inland sites, only Thiridine Point and Stonewall yielded results comparable with those from the coastal cliff sections.

4. An unconformity test

From the above it is clear that the base of the pebble marker separates strata, which carry reversed or complex magnetizations, from those which carry normal magnetizations. This is illustrated in Fig. 3a–f which contrasts NRM directions from specimens below the pebble marker (on the left), and NRM directions from specimens above and within the pebble marker (on the right). The multicomponent nature of the former is clearly reflected by the larger degree of scatter compared to the close grouping of the latter. The sandstone types above and below the marker horizon are indistinguishable in colour, mineralogies and average grain size. The distinctive change in character of the remanence at the base of a marker horizon is good evidence that the remanence is original, and constitutes an Unconformity Test first proposed by Kirschvink (1978).

The NRM results from Thiridine Point and Stonewall have been combined since they were not as densely sampled as were the coastal sections. These are plotted in Fig. 3f and reveal a similar picture to that described above, although some samples below the pebble horizon apparently contain relatively large normal components and are not greatly different from those above the marker.

The scattered directions from below the pebble marker may be explained by their acquisition, in part, after the field changed direction from reversed to normal i.e. the magnetization of the Tumblagooda Sandstone was a protracted process similar to that of many other red beds. It is widely held that red beds often possess a diagenetic chemical remanent magnetization that may have evolved over a sufficiently long period of time and involves field reversals (Roy and Park, 1974; Larson and Walker, 1975; Turner, 1980).

Overall, the magnetization of the Tumblagooda Sandstone, and its relationship to the pebble marker (Fig. 3) supports the concept of the marker horizon representing an erosional surface. Erosion could conceivably strip through a time-plane marking the geomagnetic polarity reversal. This would explain the coincidence of the base of the marker with the change in remanence polarity without appealing to the simplistic, but dubious, synchrony of the marker formation and the geomagnetic field reversal. Our preferred sequence of events is as follows.

(1) Deposition of medium to fine grained laminated and cross-bedded sandstone; (2) geo-

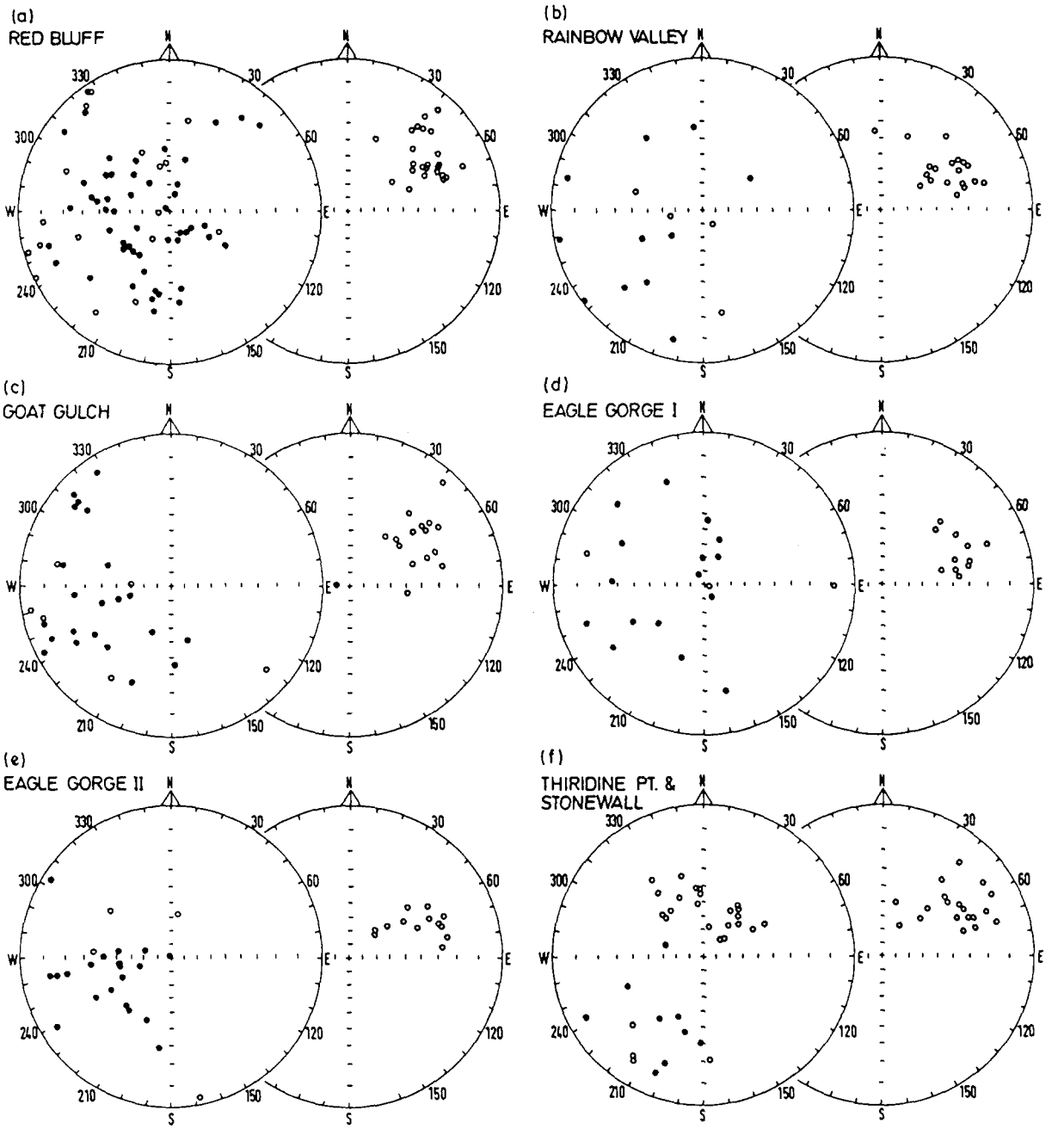


Fig. 3. Stereographic projections of NRM directions from samples above and within the pebble marker on the right, and from samples below the pebble marker on the left. Solid (open) symbols refer to the lower (upper) hemisphere. It should be noted that the complex magnetizations from below the marker are scattered compared to the simple normal magnetization above and within the marker.

magnetic transition from reversed to normal, as deposition continues (it should be noted that the magnetization is not complete at the time of deposition, so for some strata below this time-plane the magnetization will comprise both polarities); (3) continued deposition for an unspecified period;

(4) cessation of deposition; (5) erosion/stripping of depositional surface, at non-uniform rates (higher at the coastal sections than at the Thiridine Point/Stonewall sections); (6) formation of pebbly marker horizon, probably by accumulation of a pebble/granule lag on the erosion surface; (7) renewed deposition and magnetization within a period of normal geomagnetic polarity.

The non-uniform stripping is required to account for the differing proportions of normal and reversed magnetizations found immediately below the marker horizon at the coastal sections as compared to the Thiridine Point and Stonewall sections. At these latter sections a much greater proportion of normal magnetization is present suggesting less stripping has occurred at these places.

5. Implications for Gondwana apparent polar wander

From the Unconformity Test (Kirschvink, 1978) described above, the magnetization of the Tumblagooda Sandstone may be considered to date from the time of its formation. The geological age of the Tumblagooda Sandstone is apparently well constrained by older basement mineralization dates of 435 Ma, and the conformably overlying Upper Silurian Dirk Hartog Formation. On this basis we could therefore assign an age of about 420 Ma to the Tumblagooda Sandstone.

The magnetic characteristics of the Tumblagooda Sandstone within and above the marker horizon fit the Type A category of Turner (1980), in which NRM acquisition has been relatively rapid, say within 10^6–10^7 a, but long enough to average most secular variation. Remanent magnetizations of Type A red beds are excellent geomagnetic recorders and, coupled with the above plus the certain knowledge of the palaeohorizontal, the palaeomagnetic pole listed in Table 1 satisfies all the criteria set by Briden and Duff (1981) to be classified as an A pole. Note the excellent agreement in mean direction of magnetization between the sections, indicating time-averaging of the magnetization direction. Furthermore, since the Tumblagooda Sandstone is remote from the suspect terranes of the Lachlan Foldbelt of eastern Australia, this pole should be an excellent Silurian pole, not only for cratonic Australia, but also for the whole of Gondwanaland.

Although there are no other Silurian poles from Australia that qualify for A rating, the A pole from the Snowy River Volcanics is thought to be an Early Devonian pole and may even be Late Silurian. However, the Snowy River Volcanics form part of the suspect terranes of eastern Australia, that may have been rotated with respect to the craton. Indeed, a cursory comparison of the Tumblagooda Sandstone and the Snowy River Volcanics pole positions (Fig. 4b) shows them to be in striking contrast. Moreover, the difference cannot be explained by a simple rotation about a vertical axis, but suggests a large palaeolatitudinal variation (above that due to geographic location). Superficially, this would seem to prove the case for there being allochthonous (displaced) terrains present in the Lachlan Foldbelt.

If the data plotted in Fig. 4a were all that were available we would be led to the unavoidable conclusion that data from the Palaeozoic of eastern Australia are unreliable in terms of Gondwana apparent polar wandering. When the Australian data are compared to those from Africa however, it seems to be the Tumblagooda Sandstone pole that is at variance. The recent result of Hargraves et al. (1987) from the Silurian (435 Ma) Nigerian ring complexes (Air pole) must be considered to be the most reliable Palaeozoic pole for Africa available, yet it is clearly divergent from the Tumblagooda Sandstone pole, when they are plotted in the Gondwanaland reference frame (Fig. 4b). In fact the Air pole most closely agrees with the Snowy River Volcanics pole and, although there may be up to 40 Ma difference in their ages, their proximity suggests that writing-off eastern Australian pole positions may be premature. While we would not advocate the uncritical acceptance of poles from the Lachlan Foldbelt, we feel that until rotations are proven to have occurred, poles from the Lachlan Foldbelt should be accepted at face value.

Where conflicts arise the most parsimonious solution should be sought. In the present paradox, the weakest link seems to be the lower age limit

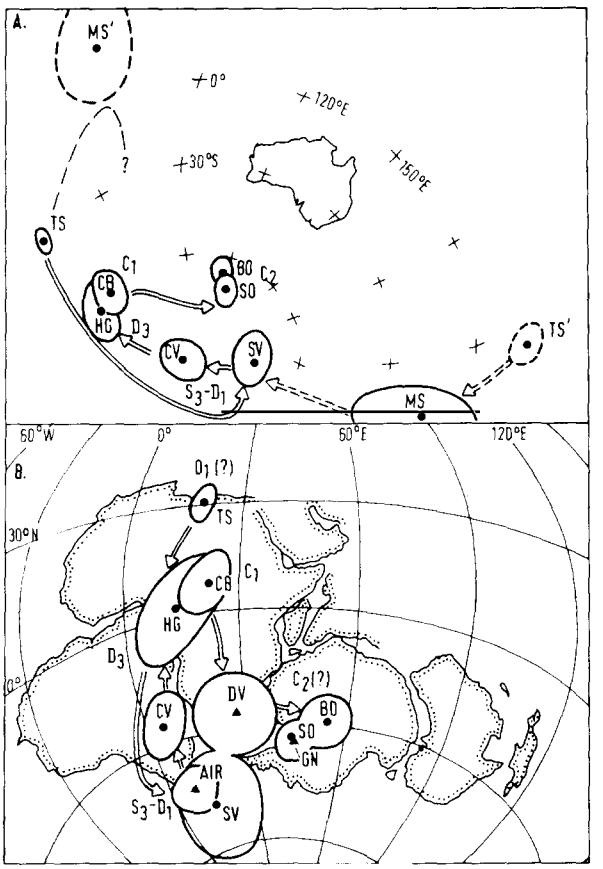


Fig. 4. High quality poles (A and B category, Briden and Duff, 1981) from the Ordovician to the Carboniferous of, (a) Australia, (circles) and (b) Africa, (triangles). Pole mnemonics are: TS, Tumblagooda Sandstone (this study); MS, (new) Mereenie Sandstones (Li et al., 1989, new data); TS′ and MS′, their anti-poles; SV, Snowy River Volcanics (Schmidt et al., 1987); CV, Comerong Volcanics (Schmidt et al., 1986); AIR, Niger Ring complexes (Hargraves et al., 1987); HG, Hervey Group (Li et al., 1990); CB, Canning Basin limestones (Hurley and Van der Voo, 1987); DV, Dwyka Varves (McElhinny and Opdyke, 1968); GN, Gneira Sandstone (Kent et al., 1984); SO, Snowy R. overprint and BO, Buchan Caves Limestone overprint (Schmidt et al., 1987).

for the Tumblagooda Sandstone. This is based on a Rb/Sr model age (435 Ma) for mineralization in basement dykes (J. Blockley, personal communication, in Hocking et al., 1987), and the assumption, in view of the lack of evidence to the contrary, that the mineralization pre-dates the Tumblagooda Sandstone (Hocking et al., 1987). Since the Tumblagooda Sandstone is cratonic, and the palaeomagnetic results are technically reliable, we are led to conclude that the Rb/Sr age must be erroneous. That it is invalid to use the Rb/Sr results to infer an age is further examined and discussed by Schmidt and Hamilton (1990). The only other possibility is that the mineralization post-dates the Tumblagooda Sandstone, but this fails to explain why the Tumblagooda and Air poles are so divergent, or why the sandstone shows no sign of mineralization. Moreover, Embleton and Schmidt (1985) report K–Ar ages of 750–550 Ma from the mineralized dolerite dykes. These same dykes yield a palaeomagnetic pole consistent with the older K–Ar ages and there is no sign of a Silurian magnetic overprint that may be expected to have accompanied the mineralizing event.

We therefore believe that the Tumblagooda Sandstone is significantly older than the Silurian, and to be consistent with palaeomagnetic results from elsewhere, both in Australia and other Gondwana continents, it would seem that the Tumblagooda Sandstone is at least as old as Early Ordovician. Since the palaeomagnetic results pertain to the stratigraphically highest beds, the base of the Tumblagooda Sandstone may easily extend into the Cambrian. The apparent conformity with the overlying Upper Silurian Dirk Hartog Formation, only observed in drill core, may be a disconformity, although this possibility has not been investigated.

Although we have been able to show that the age of deposition and the age of magnetization of the Tumblagooda Sandstone are very close, in view of the uncertainty regarding the absolute age of the Tumblagooda Sandstone, we are forced to relegate its palaeomagnetic pole to the B class status of Briden and Duff (1981). Therefore, while this pole cannot be used to calibrate the Gondwana pole path, it nevertheless represents an important definition point.

If a Cambro/Ordovician age is assumed for the Tumblagooda Sandstone, it is apparent that Gondwanaland underwent a large amount of latitudinal movement between this time and the Silurian (represented by the Air and the Snowy River Volcanic poles in Fig. 4a). As vast as Gondwanaland was (representing about 15 percent of the Earth's surface), parts of it appear to

have moved through over 60° of latitude in only 30 Ma (over 20 cm year^{-1}). While this is not necessarily identical with the spreading (half-) rates of adjoining oceans, it seems unavoidable to conclude that they were comparable. A younger age for the Tumblagooda Sandstone exacerbates this problem. These conclusions have enormous consequences for extending plate tectonic theory back in time. It would appear that plate motions in the Palaeozoic are not necessarily constrained to the relatively sedate motions of today's tectonic plates.

We also note that the gap between the Air and Snowy River Volcanics pole positions and the Tumblagooda pole position is too large to extrapolate the polarity convention backwards in time with confidence. This implies that an alternative pole path may be constructed for Gondwanaland using the anti-pole of the Tumblagooda Sandstone (Fig. 4a). Such a possibility was forseen by Schmidt and Morris (1977), but has not been seriously considered by palaeogeographers and tectonicists. In a recent study of the Amadeus Basin, however, new results from the Mereenie Sandstone yield a pole position at 16° N, 63° E, which, using the conventional polarity assignment requires even greater apparent polar wander path (APWP) for the period from the Ordovician to the Devonian. This study was instigated because the Amadeus Basin is cratonic i.e. remote from eastern Australia. Using the anti-pole of this new result from the Mereenie Sandstone and the anti-pole of the Tumblagooda Sandstone, ameliorates the spreading rate problem, but does not solve it. A longer discussion of the polarity option and the new results from the Amadeus Basin is given by Li et al. (1990). It should be mentioned that the palaeoclimate evidence of Caputo and Crowell (1985) supports the currently accepted polarity convention, with glacial centres migrating across western Gondwanaland from north to south, then north again, from the Ordovician to the Carboniferous. The alternative polarity option would place western Gondwanaland in high palaeolatitudes during the Ordovician and the Late Devonian and Carboniferous, but in low palaeolatitudes during the Silurian and Early Devonian.

Other recently acquired Devonian pole positions (Schmidt et al., 1986; Hurley and Van der Voo, 1987; Li et al., 1988) suggest that rapid APWP of Gondwana was not restricted to the Ordovician/Silurian and another episode occurred during the Devonian. The close proximity of the Air pole (435 Ma) and the Snowy River Volcanics pole (~ 400 Ma) suggests a hiatus between these periods of rapid latitudinal movement (of course any longitudinal movement would not be detected). The outcome of the debate on the age of the Canning Basin limestone magnetization (Hurley and Van der Voo, 1987; Schmidt et al., 1987) will not change this picture greatly, since the pole position in question is well defined and the pole path must pass nearby at sometime. The magnetization cannot be older than the rock, which is well dated as Late Devonian (Frasnian–Famennian) and is the preferred magnetization age of Hurley and Van der Voo, while the younger age of Early Carboniferous (Schmidt et al., 1987) decreases the rate only slightly. This Early Carboniferous timing for the pole in central Africa is favoured by the Caputo and Crowell (1985) analysis of glacial centres as they advanced across Gondwanaland. Nevertheless, there appears to be no way to avoid the conclusion that the Palaeozoic was an Era of much greater sea-floor spreading rates than those observed today.

Recent palaeogeographic reconstructions of Occidental Laurasia (Laurasia less the Orient) and Gondwanaland, based on updated Palaeozoic pole paths, show a wide palaeo-Atlantic closed in the Silurian and reformed in the Devonian (Hargraves et al., 1987; Van der Voo, 1988). This waxing and waning coincides with rapid drift phases of Gondwanaland, and the Caledonian/Acadian Orogeny coincides with the Siluro-Devonian hiatus. A similar event in the Carboniferous appears to be responsible for the Hercynian/Alleghanian Orogeny and lends considerable support for the notion of Wilson Cycles (Wilson, 1966). Hargraves et al. (1987) and Van der Voo (1988) discuss, at length, the palaeoclimatological and biogeographical implications of the updated Palaeozoic Gondwana pole path and we will not repeat them here, except to note that, apart from the considerable problem of the 20 cm year^{-1} spreading rates implied for Palaeozoic oceans, some palaeoclimatological and palaeobiogeo-

graphical objections to palaeogeographical propositions in the past appear to have been surmounted only to be supplanted with others. More, high quality Early–Mid Palaeozoic palaeomagnetic data are required from the Gondwana continents.

Acknowledgements

The field work was undertaken with the kind permission of the National Parks Authority of Western Australia. Mark Huddleston lent valuable support in the field and the laboratory. Roger Hocking (Geol. Surv. W.A.) is thanked for his discussions on the geology, and for collecting the samples from north of Kalbarri with the assistance of Mike Middleton. Don Emerson (University of Sydney) is sincerely thanked for lending his portable magnetometer for the fieldwork. The diligent refereeing by Dr. Juan Vilas was much appreciated.

References

Briden, J.C. and Duff, B.A., 1981. Pre-Carboniferous paleomagnetism of Europe north of the Alpine orogenic belt. In: M.W. McElhinny and D.A. Valencio (Editors), Paleoreconstruction of the Continents. Am. Geophys. Union Geodynamics Series, Vol. 2, pp. 178–211.

Caputo, M.V. and Crowell, J.C., 1985. Migration of glacial centers across Gondwana during the Paleozoic Era. Geol. Soc. Am. Bull., 96: 1020–1036.

Collinson, D.W., 1974. The role of pigment and specularite in the remanent magnetism of red sandstones. Geophys. J.R. Astron. Soc., 38: 253–264.

Condon, M.A., 1965. The geology of the Carnarvon Basin, Western Australia, 1, Pre-Permian stratigraphy. Bur. Miner. Resour. Geol. Geophys. Aust. Bull., 77, 78 pp.

Embleton, B.J.J. and Giddings, J.W., 1974. Late Precambrian and Lower Palaeozoic Palaeomagnetic results from South Australia and Western Australia. Earth Planet. Sci. Lett., 22: 355–365.

Embleton, B.J.J. and Schmidt, P.W., 1985. Age and significance of magnetizations in dolerite dykes from the Northampton Block, Western Australia. Australian J. Earth Sci., 32: 279–286.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. Ser. A, 217: 295–305.

Graham, J.W., 1949. The stability and significance of magnetism in sedimentary rocks. J. Geophys. Res., 54: 131–167.

Hargraves, R.B., Dawson, E.M. and Van Houten, F.B., 1987. Palaeomagnetism and age of mid-Palaeozoic ring complexes in Niger, West Africa, and tectonic implications. Geophys. J.R. Astron. Soc., 90: 705–729.

Hocking, R.M., 1978. Sedimentology of the Tumblagooda Sandstone (Silurian) in the lower Murchison River area, Western Australia: a preliminary interpretation. West. Aust. Geol. Surv. Ann. Rep., pp. 40–44.

Hocking, R.M., 1981. The Tumblagooda Sandstone, Western Australia: its type section and sedimentology. West. Aust. Geol. Surv. Ann. Rep., 1980, pp. 53–61.

Hocking, R.M., Moors, H.T. and van der Graff, W.V.E., 1987. The Geology of the Carnarvon Basin, Western Australia. Geol. Surv. W.A. Bull., 133, 289 pp.

Hurley, N.F., and Van der Voo, R., 1987. Paleomagnetism of Upper Devonian reefal limestone, Canning Basin, Western Australia. Geol. Soc. Am. Bull., 98: 138–146.

Kent, D.V., Dia, O. and Sougy, J.M., 1984. Paleomagnetism of lower-middle Devonian and upper Proterozoic-Cambrian(?) rocks from Mejeria (Mauritania, West Africa). In: R. van der Voo, C.R. Scotese and N. Bonhommet (Editors), Plate Reconstructions from Paleozoic Paleomagnetism. AGU Geodynamics Series, 12, pp. 99–115.

Kirschvink, J.L., 1978. The Precambrian-Cambrian boundary problem: paleomagnetic directions from the Amadeus Basin, Central Australia. Earth Planet. Sci. Lett., 40: 91–100.

Konecki, M.C., Dickens, J.M. and Quinlan, T., 1958. The geology of the coastal area between the Lower Gascoyne. and Murchison Rivers, Western Australia. Bur. Miner. Resour. Geol. Geophys. Aust. Rep., 37, 144 pp.

Larson, E.E. and Walker, T.R., 1975. Development of chemical remanent magnetization during early stages of red bed formation in Late Cenozoic sediments, Baja California. Geol. Soc. Am. Bull., 86: 639–650.

Li, Z.X., Schmidt, P.W. and Embleton, B.J.J., 1988. Palaeomagnetism of the Hervey Group, central New South Wales, and its tectonic implications. Tectonics, 7: 351–367.

Li, Z.X., Powell, C.McA., Embleton, B.J.J. and Schmidt, P.W., 1990. New palaeomagnetic results from the Amadeus Basin and their implications for stratigraphy and tectonics. In: R.J. Korsch (Editor), The Geology and Geophysics of the Amadeus Basin, Central Australia, BMR Special Vol., in press.

Livermore, R.A., Smith, A.G. and Briden, J.C., 1985. Palaeomagnetic constraints on the distribution of the continents in the Late Silurian and Early Devonian. Philos. Trans. R. Soc. London, Ser, B: 309: 29–56.

McElhinny, M.W. and Opdyke, N.D., 1968. Paleomagnetism of some Carboniferous glacial varves from Central Africa. J. Geophys. Res., 67: 53–58.

McFadden, P.L. and Jones, D.L., 1981. The fold test in palaeomagnetism. Geophys. J.R. Astron. Soc., 76: 53–58.

Miller, J.D. and Kent, D.V., 1988. Paleomagnetism of the Siluro-Devonian Andreas redbeds: evidence for an Early Devonian supercontinent. Geophys. Res. Lett., 16: 195–198.

Roy, J.L. and Park, J.K., 1974. The magnetization process of

certain red beds: vector analysis of chemical and thermal results. Can. J. Earth Sci., 11: 437–471.

Schmidt, P.W., 1982. Linearity spectrum analysis of multi-component magnetizations and its application to some igneous rocks from south-eastern Australia. Geophys. J.R. Astron. Soc., 70: 647–665.

Schmidt, P.W. and Embleton, B.J.J., 1976. Palaeomagnetic results from sediments of the north Perth Basin, and their bearing on the timing of regional lateritization. Palaeogeogr., Palaeoclimatol., Palaeoecol., 19: 257–273.

Schmidt, P.W. and Morris, W.A., 1977. An alternative view of the Gondwana Paleozoic apparent polar wander path. Can. J. Earth Sci., 14: 2674–2678.

Schmidt, P.W. and Hamilton, P.J., 1990. Palaeomagnetism and the age of the Tumblagooda Sandstone, Western Australia. Austral. J. Earth. Sci., in press.

Schmidt, P.W., Embleton, B.J.J., Cudahy, T.J. and Powell, C.McA., 1986. Prefolding and premegakinking magnetizations from the Comerong Volcanics, New South Wales, Australia, and their bearing on the Gondwana pole path. Tectonics, 5: 135–150.

Schmidt, P.W., Embleton, B.J.J. and Palmer, H.C., 1987. Pre- and post-folding magnetizations from the Early Devonian Snowy River Volcanics and Buchan Caves Limestone, Victoria. Geophys. J. R. Astron. Soc., 91: 155–170.

Turner, P., 1980. Continental Red Beds. Dev. in Stratig., 29: 562 pp.

Van der Voo, R., 1988. Paleozoic paleography of North America, Gondwana, and intervening displaced terranes: comparisons of paleomagnetism with paleoclimatology and biogeographical patterns. Geol. Soc. Am. Bull., 100: 311–324.

Wilson, J.T., 1966. Did the Atlantic close and then re-open? Nature, 211: 676–681.

Zijderveld, J.D.A., 1967. A.C. demagnetization in rocks. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (Editors), Methods in Palaeomagnetism. Elsevier, New York, pp. 254–286.